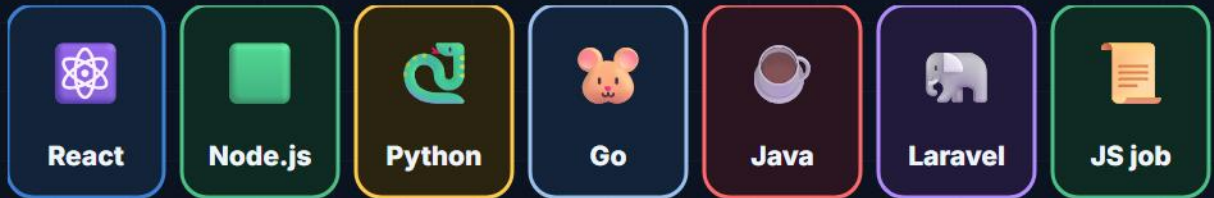


7 stacks, **one way** to ship them

React, Node.js, Python, Go, Java, Laravel, JavaScript:
Dockerfile → Compose → Kubernetes.



THE USUAL WAY

7 stacks · 7 deploy recipes · YAML copied, not understood



THIS LAB

one contract · one image per app · one Compose file
the same images on Kubernetes, the right resource each

Build → Dockerize → Compose → Deploy → Troubleshoot

THE PAIN

Where multi-stack deploys **go wrong**

every stack seems to need its own magic.



7 stacks, 7 recipes

npm, pip, go, Maven, Composer:
each team ships differently



Copied YAML

selector, targetPort, probes:
pasted, never understood



Compose → K8s guess

depends_on, .env, volumes:
"converted" by trial and error



No failure practice

the first CrashLoopBackOff
happens in production

THE IDEA

Think **shipping containers**

seven different factories, one standard box, one port with one crane



7 factories = 7 languages and frameworks



one standard box

image · PORT · /health · /ready · env · stdout



one port, one crane

Kubernetes moves boxes, not languages

learn the box once, ship any stack

WHO IT IS FOR

Use it when... skip it when...

a beginner lab, not a production blueprint

✓ USE IT WHEN YOU...

-  can run one container, not a platform
-  have a Compose file and must move to Kubernetes
-  ship services in several languages
-  want to debug CrashLoopBackOff calmly
-  prepare for a DevOps / platform interview

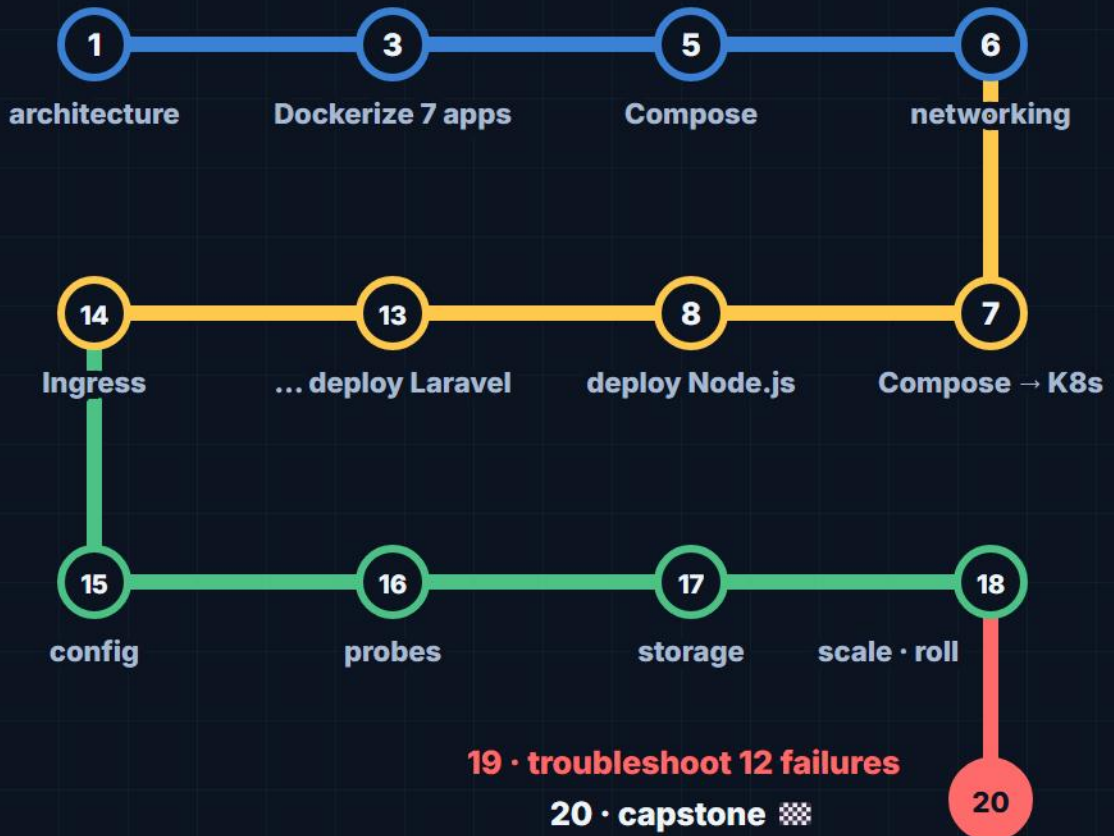
⏮ SKIP IT WHEN YOU...

need a production database setup (backups, replicas)
already run Helm, GitOps and autoscaling daily
(the lab says so itself: single-replica PostgreSQL is for learning)

HOW YOU USE IT

A metro map in 20 stops

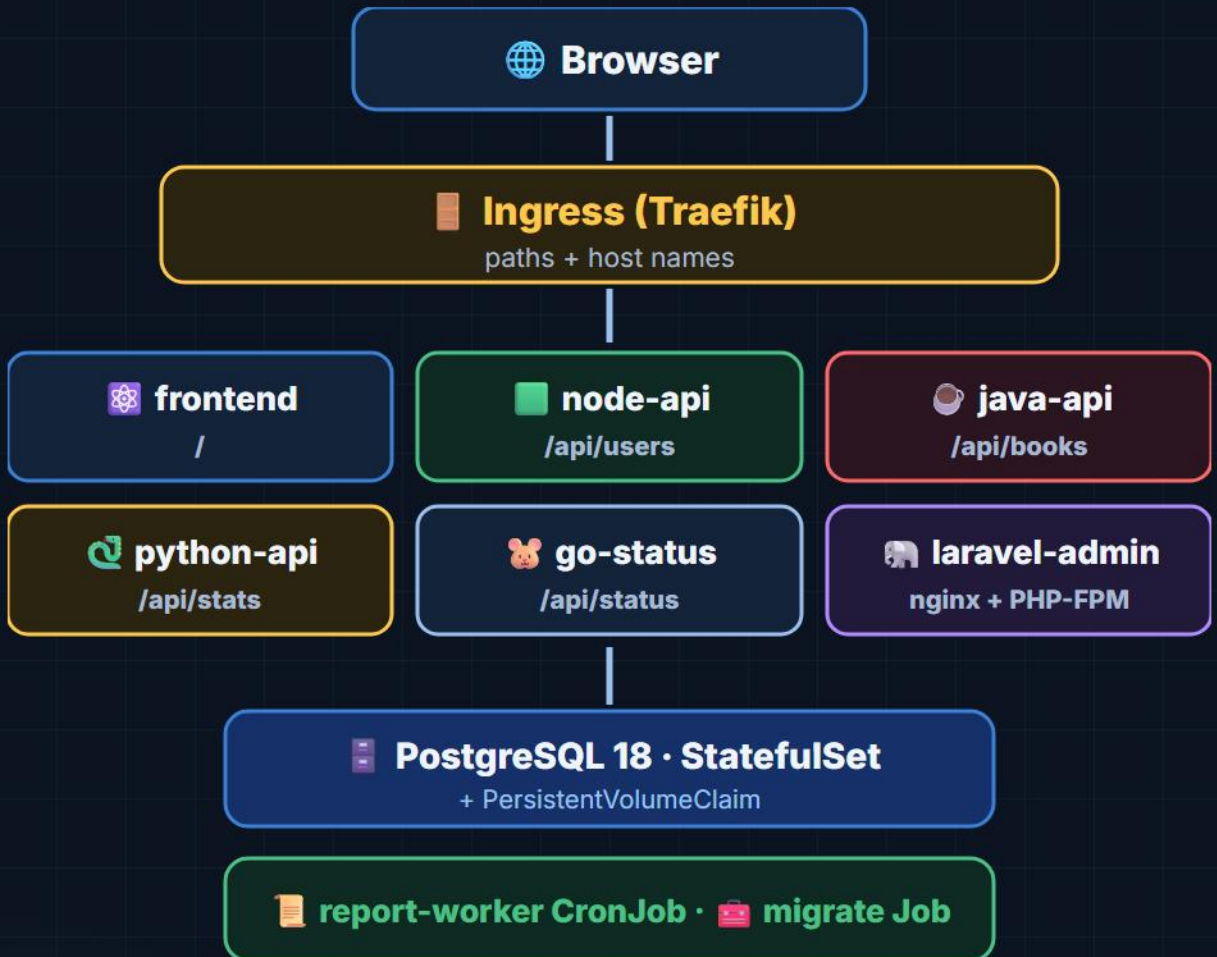
each stop ends with something that works



ARCHITECTURE

One Ingress, six services

the Bookshop platform on Kubernetes



HOW IT WORKS

One contract, every command run

the lessons are tested Markdown, not screenshots

THE CONTRACT (all 7 apps)

🔧 PORT from env

⚙️ config from env

📄 logs to stdout

💚 /health = alive

🟢 /ready = can serve

🧘 non-root user



lesson.md



mdrun.py runs it



real output back

GitHub Actions, on every change

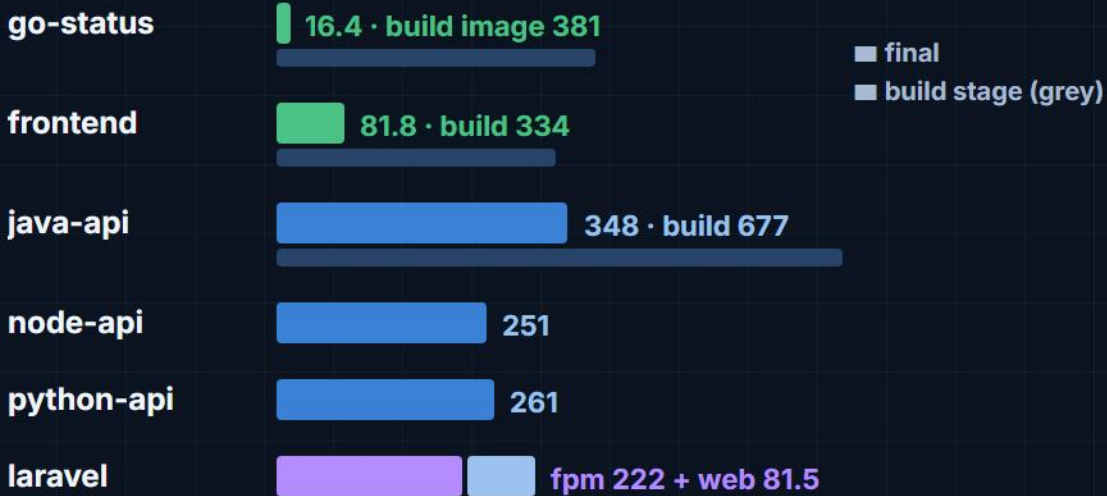
🐳 7 apps on Docker · 🧱 the platform on Compose

⚙️ kind cluster: lessons, 12 failures, labs, capstone

MEASURED RESULTS · LAB

Real sizes, real errors

image size, final vs build stage (MB)



4.5 s

Spring Boot start in the cluster

22 checks

capstone/verify.sh

```
ImagePullBackOff                                OOMKilled · exit 137
CrashLoopBackOff: "Cannot find module"
503 "no available server"
"password authentication failed"
```

STUDY MATERIAL

Everything to learn it properly

free, in one repository

7

app lessons

15

Kubernetes lessons

15

concept docs

12

troubleshooting labs

6

challenge labs

14

tutorial chapters



Compose lesson · capstone with an on-call scenario



36-chapter video · full + silent version



75-page study guide PDF



79-term glossary · 25 interview questions

RUN IT YOURSELF

Six steps to the whole platform

Docker, kind, kubectrl, Helm. No cloud account.

```
1 $ git clone .../sufyan-devops-multi-stack-kubernetes
```

```
2 $ docker compose up -d --wait
```

```
3 $ kind create cluster --config ...
```

```
4 $ helm install traefik traefik/traefik ...
```

```
5 $ bash capstone/deploy.sh
```

```
6 $ bash capstone/verify.sh
```

bookshop.localhost:8080 ✓

every lesson shows the real output to compare

FREE AND OPEN SOURCE

Dockerize. Deploy. **Fix it.**

How many stacks does your team ship to Kubernetes? 🗨️



The repository

[github.com/sufyanahmadkamboh/
sufyan-devops-multi-stack-kubernetes](https://github.com/sufyanahmadkamboh/sufyan-devops-multi-stack-kubernetes)



All my projects

sufyanahmadkamboh.github.io

7 apps · Compose · Kubernetes · 12 failures · video · PDF